



Investigation of biodistribution by liquid-phase microextraction: Using a fatal diphenidol poisoning case

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ABSTRACT

Liquid-phase microextraction (LPME) possesses a high potential to isolate organic substances from different sample matrices. In this work, LPME was applied for the first time to investigate the biodistribution of diphenidol in different biofluids, organs, and brain regions using a fatal poisoning case. Since the LPME of diphenidol hasn't been reported, the effect of supported liquid membrane (SLM), acceptor and donor phases, and extraction time on LPME performance was investigated first. The solvents of 2-nonanone and 2-nitrophenyl octyl ether (NPOE) were found to be stable and efficient SLMs for LPME of diphenidol from biofluids and tissue samples, respectively. At steady state, the LPME recoveries for different sample matrices were in the range of 87 %–91 %. Due to the clean-up capability of LPME and the relatively high concentration of diphenidol in the fatal poisoning case, the proposed LPME systems were validated with related sample matrices using HPLC-UV for the determination. The methods displayed good linearity ($R^2 \geq 0.9943$), and the limits of detection were 0.30 mg L^{-1} , 0.28 mg L^{-1} , and $2.7 \text{ } \mu\text{g g}^{-1}$ for blood, urine, and liver samples, respectively. Meanwhile, the precision ($\leq 13\%$), accuracy (90–110%), and matrices effect ($\pm 15\%$) were satisfactory at low, medium, and high concentrations. In addition, the stability, carryover, and dilution integrity met the requirements of ASB Standard 036. Finally, the proposed method was successfully applied to evaluate the biodistribution of diphenidol in five different biofluids, five organs, and six brain regions from a fatal poisoning case. Generally, the distribution of diphenidol in biofluids was lower than that in the organs and brain regions, and the highest concentration of diphenidol was observed in the liver, which is very important for the selection of inspection samples in forensic toxicological analysis. Therefore, LPME was proved to be a powerful tool for the investigation of biodistribution and postmortem redistribution in the fields of forensics.

1. Introduction

Diphenidol is a non-phenothiazine antiemetic drug that suppresses symptoms such as nausea and vomiting by modulating the vestibular nervous system [1,2]. As a result of its anti-vertigo properties, diphenidol is widely used to treat Meniere's disease and labyrinthine disease [3]. Nevertheless, the overdose of diphenidol may lead to dry mouth, blurred vision, headache, palpitations, fatigue, and, in severe cases, hallucinations, loss of consciousness, and even death [4,5]. In clinical and forensic practice, there has been a gradual increase in cases of poisoning associated with diphenidol due to its widespread use [6,7].

For the determination of diphenidol in biological samples, gas chromatography (GC) [8], high-performance liquid chromatography with ultraviolet (HPLC-UV) [4], gas chromatography-mass spectrometry (GC-MS) [1,7], as well as liquid chromatography-mass spectrometry (LC-MS) [6,9,10] have been developed. The methods of GC-MS and LC-MS possess high sensitivity and specificity, but the instrumentation is expensive. HPLC-UV is widely used in the quantification of poisons in fatal poisoning cases because of its convenience. However, due to the complex sample matrix in biological samples, sample pretreatment was essential prior to instrumental analysis [11]

Liquid-liquid extraction (LLE) is a commonly used pretreatment

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method for diphenidol [1,4,6,8,10,12]. For example, Liu et al. introduced a technique for quantifying diphenidol in blood and liver by LLE in combination with LC-MS [6]. Additionally, Liu et al. established a technique for detecting diphenidol in blood and urine by an approach of LLE-GC [6,8]. However, a significant drawback of LLE is the requirement for large amounts of organic solvents and its complex operation [13]. In recent decades, sample preparation has shown a development trend towards miniaturization and green.

Liquid-phase microextraction (LPME) has been developed in the 1990s as the miniaturization format of LLE [14]. The principle of (Fig. 1) LPME is based on the distribution of substances in the sample, acceptor solution, and extractant (the supported liquid membrane). Generally, for basic drugs such as diphenidol, the sample was adjusted to a pH-value of at least 2 units than the target pKa-value to make the target in a molecular state, and the acceptor solution should be acidic to ionize the target and prevent back extraction. During LPME, neutral targets will transfer from the sample into the SLM and further into the acceptor solution because of ionization. As such, LPME possesses the features of simple operation, sample clean-up capability, and tuning selectivity [15]. In addition, compared with LLE, LPME has the advantage of using fewer solvents and is more environmentally friendly [16,17]. To date, LPME has been widely used for the separation of basic drugs from blood [18,19], urine [20,21], and tissue samples [22], although LPME of diphenidol has not been reported.

In forensic science, fatal diphenidol poisoning cases happen from time to time, and explanations for possible causes of death and the mechanism of poison often rely on the distribution of poisons *in vivo* [23]. However, most of the detected biological samples were blood or urine in cases of diphenidol poisoning [1,4–7]. To date, the distribution of diphenidol in reproductive organs was reported only in a diphenhydramine overdose case using the sample preparation method of solid-phase extraction (SPE), and the concentration of diphenidol was higher in organs but below the toxic level [7]. However, comprehensive studies on the biodistribution of diphenidol in fatal diphenidol poisoning cases are still lacking.

Therefore, in this work, we aim to systematically investigate the biodistribution of diphenidol in a fatal poisoning case by LPME. In this study, the LPME of diphenidol from whole blood, urine, and tissue samples was investigated, and the developed LPME methods were validated in combination with HPLC-UV due to the high concentration of diphenidol in a fatal poisoning case. Finally, the validated method was applied to investigate the biodistribution of diphenidol in a fatal diphenidol poisoning case to assist the selection of inspection samples for fatal diphenidol poisoning cases in forensic toxicological analysis.

2. Materials and methods

2.1. Chemicals and materials

Diphenidol and formic acid were obtained from Aladdin Chemical Reagent Co. (Shanghai, China). 1-ethyl-2-nitrobenzene (ENB), 2-nitrophenyl octyl ether (NPOE), tributyl phosphate (TBP), 2-nonanone, *n*-nonanol, hydrochloric acid (HCl), phosphoric acid (H₃PO₄), formic acid (FA), trifluoroacetic acid (TFA), and acetic acid (HAC) were purchased

from Sigma Aldrich (Shanghai, China). Sodium hydroxide (NaOH) was supplied by Sinopharm Chemical Reagent Co. Ltd. (Shanghai, China). Formic acid was of chromatographic purity grade, and other chemicals were of analytical grade. The ultrapure water utilized in the experiment was procured from a Milli-Q water purification system (Molsheim, France). The flat membrane, Accurel polypropylene (PP) 1E (R/P), with an average thickness of 100 μm , was supplied by Membrana (Wuppertal, Germany).

2.2. Standard solutions and biological samples

A stock solution was prepared by dissolving diphenidol in methanol at a concentration of 1 mg mL^{-1} and stored at -20° away from light. To obtain a working solution, the stock solution was diluted to 2 $\mu\text{g mL}^{-1}$ with 10 mM NaOH and preserved at 4 $^\circ\text{C}$.

This study was authorized by the Ethics Committee of Tongji Medical College (NO. [2023] IEC(A003)). Diphenidol-free urine and whole blood samples were from the volunteers affiliated with Huazhong University of Science and Technology who had not recently consumed diphenidol. Diphenidol-free urine and whole blood samples were stored at -20° . Pork liver from a local supermarket (Wuhan, China) was used as the blank tissue sample. The real samples including biofluids (urine, heart blood, peripheral blood, pericardial effusion, and cerebrospinal), organs (liver, heart, spleen, lung, and kidney), and brain regions (frontal lobe, parietal lobe, temporal lobe, occipital lobe, cerebellum, internal capsule) were obtained from a fatal diphenidol poisoning case. The pork liver, organs and brain regions were frozen ground with 10 mM NaOH to obtain a tissue suspension of 0.1 g mL^{-1} .

2.3. LPME setups and procedures

A flat membrane-based LPME device as described elsewhere was utilized for LPME of diphenidol from biofluids, liver and tissue samples [24]. It comprises an acceptor phase compartment made of a pipette tip sealed with a flat membrane (SLM support), and the donor phase container (a 2 mL Eppendorf PP tube). 5 μL of SLM solvent is applied on the SLM support at the bottom of the acceptor phase compartment, and then the acceptor phase (200 μL acidic solution) is filled into the acceptor phase compartment. Meanwhile, 800 μL diluted sample (whole blood, urine, or tissue suspension) with 10 mM NaOH (1:1, v/v) was added into the donor phase container. Finally, the acceptor phase compartment is inserted into the donor phase container and LPME is initiated by turning on the constant temperature mixer (Hangzhou MIULAB Instrument Co. Ltd., Hangzhou, China). After the desired extraction time, the acceptor solution was collected and analyzed using HPLC-UV.

2.4. HPLC-UV

The detection of the analytes was conducted by an Agilent 1260 Infinity II LC System (Agilent Technologies, CA, USA), with a Quaternary Pump VL G7111A. Poroshell 120 EC—C18 column (150 $\text{mm} \times 4.6$ mm, 4 μm) was used to separate analytes at 45 $^\circ\text{C}$ in a column oven of Multicolumn Thermostat G7116A. The UV–vis detector of Variable Wavelength Detector G7114A was operated at 214 nm, and the injection volume by Vialsampler G7129A was 10 μL . Mobile phase A was 0.1 % (v/v) formic acid solution containing 5 % methanol (v/v), and mobile phase B was methanol containing 5 % formic acid solution (0.1 %, v/v), which was controlled by a quaternary pump. Baseline separation of diphenidol was achieved using 57 % B and 43 % A at a flow rate of 0.8 mL min^{-1} .

2.5. Calculations

The calculation of diphenidol recovery was performed with the following equation:

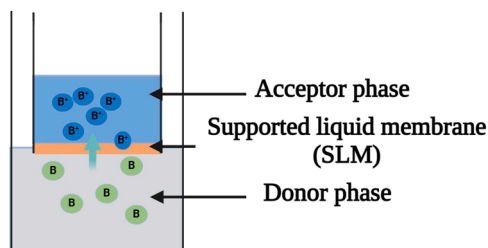


Fig. 1. Schematic representation of LPME principle.

$$\text{Recovery} = \frac{C_A \times V_A}{C_D \times V_D} \times 100\% = \frac{C_A}{4C_D} \quad (1)$$

Here, C_D and C_A represent the initial concentration of the analyte in the donor phase and the concentration of the analyte in the acceptor phase after extraction, respectively. V_A and V_D represent the volumes of the acceptor solution (200 μL) and the sample (800 μL).

The matrix effect (ME) on the method was calculated by the following equation:

$$\text{ME} = \frac{A_m}{A_{\text{std}}} \times 100\% \quad (2)$$

Where A_m and A_{std} were the peak areas of the spiked acceptor solution after LPME from the blank biological sample and the spiked pure acceptor solution, respectively.

3. Results and discussion

3.1. Optimization of LPME

3.1.1. SLM solvent

SLM solvent plays a key role in LPME and significantly affects the recovery of diphenidol. According to the criteria of SLM solvent, organic solvents must meet certain criteria including insolubility with the aqueous phases, strong affinity to the SLM support and target analytes, and good stability [25]. Therefore, ENB, NPOE, TBP, 2-nonanol, and *n*-nonanol were investigated (Fig. 2). The results showed that the LPME recovery of diphenidol was dependent on sample matrices, which was consistent with previous observations [26,27]. LPME using the SLM of 2-nonanol provided the highest recoveries for blood and urine samples, while NPOE was superior for the tissue sample. In addition, the chromatograms of the acceptor solution after LPME of spiked biological samples using 2-nonanol or NPOE as the SLM displayed no additional peaks (Fig. 3), which indicated that LPME possesses high selectivity and efficient sample clean-up capability. Thus, in the following experiments, 2-nonanol and NPOE were selected for the LPME of diphenidol from biofluids (whole blood and urine) and tissue samples.

3.1.2. Acceptor phase

In LPME, the acceptor phase should be acidic to facilitate the

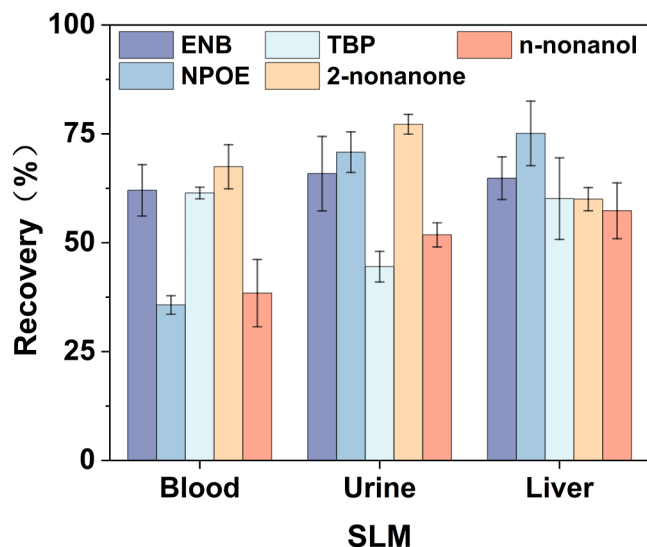


Fig. 2. Effect of SLM solvent. Extraction conditions: SLM solvent: 5 μL ; donor phase: 800 μL spiked whole blood, urine, and liver diluted with 10 mM NaOH (1:1, v/v); acceptor phase: 200 μL HCl (10 mM); extraction time: 30 min; stirring speed: 1000 rpm.

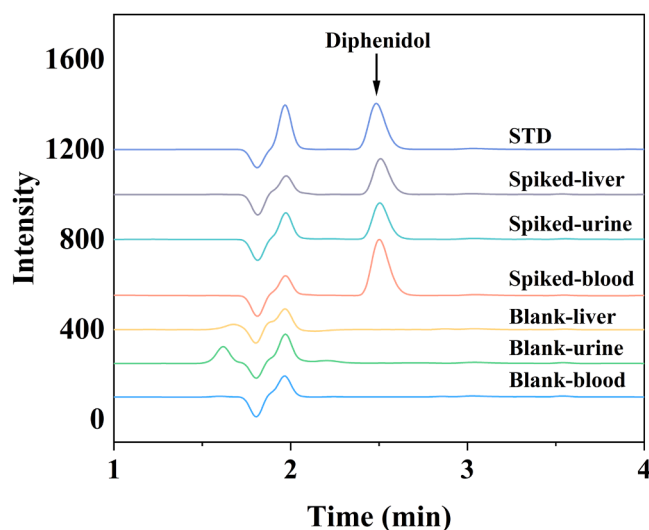


Fig. 3. HPLC-UV chromatograms of working solution (STD), LPME extracts from blank samples (blood, urine, and liver) and spiked samples (blood, urine, and liver). LPME conditions: SLM solvent: 5 μL 2-nonanol (blood and urine) or NPOE (liver); donor phase: 800 μL spiked whole blood, urine, and liver diluted with 10 mM NaOH (1:1, v/v); acceptor phase: 200 μL HAC (blood and urine) or HCl (liver); extraction time: 45 min; stirring speed: 1000 rpm.

ionization of diphenidol and prevent it from re-entering the SLM or sample. 10 mM HCl, H_3PO_4 , HAC, FA, and TFA were tested to optimize the background count ions in the acceptor phase. The impact of different acids is shown in Fig. 4a, and all tested acids provided efficient LPME. Among them, HAC was superior for both blood and urine samples, and HCl and FA showed higher recoveries for the liver samples. Consequently, combined with previous experience, HAC (for blood and urine) and HCl (for liver) were selected as the optimal acceptor phase solution, respectively. In addition, the pH of the acceptor solution directly impacts the efficiency and selectivity of LPME. Thus, the acceptor solutions (HAC and HCl) at concentrations ranging from 0.1 mM to 100 mM were tested. It was found that a higher recovery of diphenidol was obtained at a concentration of 50 mM (Fig. 4b). Therefore, the optimal acceptor phase solution was determined to be 50 mM HAC and 50 mM HCl for biofluids and liver samples, respectively.

3.1.3. Extraction time

The mass transfer of diphenidol between the donor phase, SLM, and acceptor phase is driven by passive diffusion [28]. Therefore, the extraction recovery of diphenidol was significantly affected by the extraction time. In this work, the effect of extraction time on diphenidol extraction efficiency was evaluated from 5 min to 120 min. The results (Fig. 5) indicated that the recovery of diphenidol from all three different biological samples generally increased with time until reaching the equilibrium. For urine samples, 30 min was required for the equilibrium, while 90 min and 60 min were required for whole blood and liver samples, respectively. At equilibrium, the LPME recoveries for urine, whole blood, and liver samples ranged from 87 % to 91 %.

In a word, efficient LPME of diphenidol from biological samples was achieved, and the LPME recoveries for urine, whole blood, and liver samples ranged from 87 % to 91 % at equilibrium.

3.2. Evaluation of LPME-HPLC-UV

Due to the relatively high concentration of diphenidol in the fatal poisoning case, and the high selectivity and efficient sample clean-up capability of LPME (Fig. 3), the proposed LPME systems were evaluated using HPLC-UV for the determination of diphenidol. To explore the reliability and the applicability of LPME-HPLC-UV, the matrix effect was

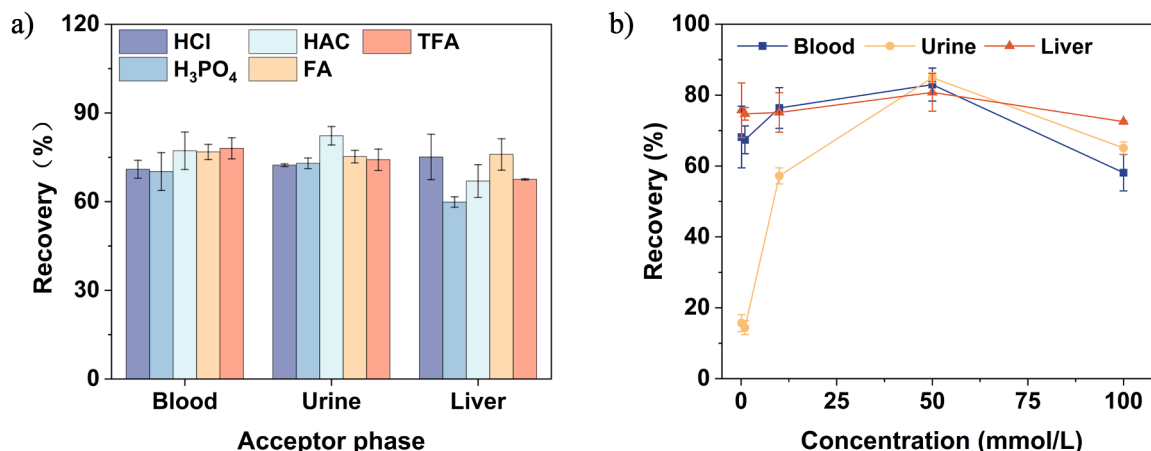


Fig. 4. Effect of acceptor phase on LPME of diphenidol: types of acids (a), the concentration of HAC (blood and urine) or HCl (liver). Extraction conditions: SLM solvent: 5 μ L 2-nonanone (blood and urine) or NPOE (liver); donor phase: 800 μ L spiked whole blood, urine, and liver diluted with 10 mM NaOH (1:1, v/v); acceptor phase: 200 μ L; extraction time: 30 min; stirring speed: 1000 rpm.

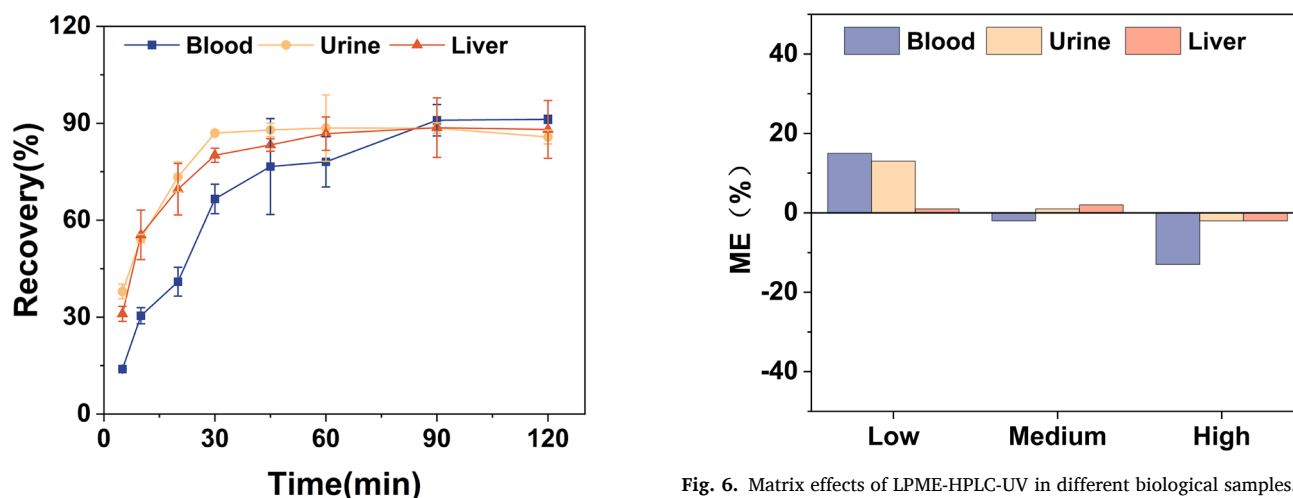


Fig. 5. Optimization of Extraction Time. Extraction conditions: SLM solvent: 5 μ L 2-nonanone (blood and urine) or NPOE (liver); donor phase: 800 μ L spiked whole blood, urine, and liver diluted with 10 mM NaOH (1:1, v/v); acceptor phase: 200 μ L HAC (blood and urine) or HCL (liver); stirring speed: 1000 rpm.

first investigated at three concentration levels: low (1 mg L^{-1} for biofluids, and 10 $\mu\text{g g}^{-1}$ for liver tissue), medium (10 mg L^{-1} for biofluids, and 100 $\mu\text{g g}^{-1}$ for liver tissue), and high (100 mg L^{-1} for biofluids, and 1000 $\mu\text{g g}^{-1}$ for liver tissue). The results are shown in Fig. 6, and the ME for the three biological samples at different concentrations was in the range of -13% to 15% , suggesting that the ME of the proposed methods was acceptable.

Subsequently, the proposed LPME-HPLC-UV methods were validated using spiked whole blood, urine, and liver samples individually. The concentrations of the spiked whole blood and urine samples were in the range of 0.5–100 $\mu\text{g mL}^{-1}$, and those for tissue samples were 5–1000 $\mu\text{g g}^{-1}$. The linear range, coefficient of determination (R^2), limit of detection (LOD, $S/N = 3$), lower limit of quantification (LLOQ, $S/N = 5$), accuracy, intraday repeatability and interday repeatability for biological samples were summarized in Table 1. The LODs for whole blood, urine, and liver were 0.30 mg L^{-1} , 0.28 mg L^{-1} , and 2.70 $\mu\text{g g}^{-1}$, respectively. Correspondingly, the LLOQ values for whole blood, urine, and liver were 0.5 $\mu\text{g L}^{-1}$, 0.5 $\mu\text{g L}^{-1}$, and 4.5 $\mu\text{g g}^{-1}$, respectively. These data indicated that the sensitivity of the current methods meets the determination of diphenidol in a lethal case [15]. The R^2 values were ≥ 0.9943 in the range of 1–100 mg L^{-1} for both whole blood and urine, and that for the

Fig. 6. Matrix effects of LPME-HPLC-UV in different biological samples. Blood and urine were spiked at 1 mg L^{-1} (low), 10 mg L^{-1} (medium), and 100 mg L^{-1} (high). Liver was spiked at 10 $\mu\text{g g}^{-1}$ (low), 100 $\mu\text{g g}^{-1}$ (medium) and 1000 $\mu\text{g g}^{-1}$ (high). Extraction conditions: SLM solvent: 5 μ L 2-nonanone (blood and urine) or NPOE (liver); donor phase: 800 μ L spiked whole blood, urine, and liver diluted with 10 mM NaOH (1:1, v/v); acceptor phase: 200 μ L HAC (blood and urine) or HCL (liver); extraction time: 45 min; stirring speed: 1000 rpm.

liver was 0.9978 in the range of 10–1000 $\mu\text{g g}^{-1}$. In addition, the accuracy and precisions intra-day and inter-day were investigated at low, medium, and high concentration levels for all sample matrices, respectively (Table 1). At all three levels, the accuracy ranged from 90 % to 110 %, and the intra-day and inter-day precisions were $\leq 10\%$ and $\leq 13\%$, respectively.

In addition, according to ASB Standard 036, the carryover and dilution integrity were evaluated. To assess the carryover, blank samples, and spiked samples at a level of 300 mg L^{-1} (whole blood and urine) and 3000 $\mu\text{g g}^{-1}$ (liver), which was as three times the highest concentration in the linear range were extracted by LPME and analyzed by HPLC-UV, and no analyte residues were detected in the blank samples (Table 2). Meanwhile, the dilution integrity was evaluated by diluting samples with concentrations of 300 mg L^{-1} (in blood and urine) and 3000 $\mu\text{g g}^{-1}$ (in liver tissue) fivefold with related blank samples. As shown in Table 2, the method had good accuracy (99–104%) and precision (1–3%) for all samples, indicating good dilution integrity.

In a word, all the results above showed that the analytical characteristics of LPME-HPLC-UV fulfill the requirements of bioanalysis. In addition, the proposed method was compared with other methods for

Table 1
Linear range, R², LOD, LLOQ precision and accuracy for LPME-HPLC-UV in whole blood and urine (n = 4).

Sample	Linear range	R ²	LOD	LLOQ	Accuracy (%)			Intra-day precision (%CV, n = 4)			Inter-day precision (%CV, n = 4)		
					Low	Medium	High	Low	Medium	High	Low	Medium	High
Blood ^{a)}	1–100	0.9943	0.30	0.5	110	90	98	3	4	7	13	8	2
Urine ^{a)}	1–100	0.9992	0.28	0.5	109	97	101	5	7	5	13	9	4
Liver ^{b)}	10–1000	0.9978	2.70	4.5	108	102	102	10	7	6	9	5	7

^{a)} The concentration for accuracy and precision are 1 mg L⁻¹ (low), 10 mg L⁻¹ (medium), and 100 mg L⁻¹ (high). The concentration unit for the linear range, LOD and LLOQ are mg L⁻¹.
^{b)} The concentration for accuracy and precision are 10 µg g⁻¹ (low), 100 µg g⁻¹ (medium), and 1000 µg g⁻¹ (high). The concentration unit for the linear range, LOD and LLOQ are µg g⁻¹.

Table 2
Carryover and dilution integrity of LPME-HPLC-UV.

Sample	Carryover (%)	Dilution integrity	
		Accuracy (%)	Precision (%CV)
Blood	3	99	1
Urine	1	102	3
Liver	1	104	3

the detection of diphenidol. As shown in Table 3, the proposed methods displayed comparable analytical performance in terms of recovery and accuracy, but required less organic solvent, and less sample amount. Therefore, LPME is applicable for biofluids and tissue samples possessing high potential for investigation of biodistribution.

3.3. Investigation of biodistribution by LPME-HPLC-UV

Finally, the validated LPME-HPLC-UV methods were used to investigate the biodistribution of diphenidol in the biofluids (urine, heart blood, peripheral blood, pericardial effusion and cerebrospinal), organs (liver, heart, spleen, lung and kidney), and brain regions (frontal lobe, parietal lobe, temporal lobe, occipital lobe, cerebellum, internal capsule) from a fatal diphenidol poisoning case. The LPME-HPLC-UV method for whole blood was used for the determination of diphenidol in the biofluids except urine, while the method for the liver was applied for organs and brain regions. The levels of diphenidol in the biofluids, major organs, and brain regions are summarized in Fig. 7. The results demonstrated that the distribution of diphenidol in biofluids was lower than that in the organs and brain regions. Meanwhile, diphenidol exhibited the highest concentration in heart blood among the various body fluids, while the liver contained the highest level of diphenidol among the measured biological samples. This might be due to the fact that diphenidol is mainly metabolized in the liver, and taking a large amount of diphenidol leads to an accumulation of diphenidol in the liver. In addition, this study also showed that the brain is one of the target organs for diphenidol poisoning. It was further demonstrated that diphenidol overdose promotes sustained H1 receptor antagonist

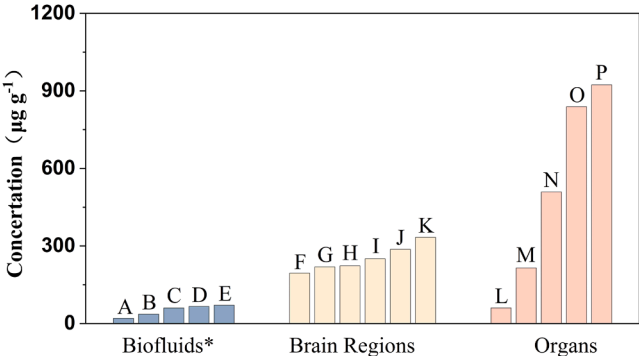


Fig. 7. Biodistribution of diphenidol in a fatal poisoning case. A-P represented cerebrospinal, urine, pericardial effusion, peripheral blood, heart blood, internal capsule, cerebellum, frontal lobe, temporal lobe, parietal lobe, occipital lobe, heart, kidney, lung, spleen, and liver. The liver was determined at a homogenate concentration of 0.02 g mL⁻¹ to ensure that the diphenidol concentrations were within the linear range.
*The concentration unit for the biofluids is µg mL⁻¹.

antagonists and anticholinergic effects in the central nervous system, resulting in impaired consciousness, convulsions, and respiratory failure [29,30].

4. Conclusion

In this work, LPME was applied for the first time to isolate diphenidol from different biological samples and further used to investigate the biodistribution of diphenidol in different biofluids, organs, and brain regions using a fatal poisoning case. Under optimal conditions, the LPME recoveries for different biological samples were in the range of 87 %–91 %. LPME-HPLC-UV displayed sufficient sample clean-up capability and negligible matrix effect. In addition, LPME-HPLC-UV showed satisfactory analytical characteristics for bioanalysis and fulfilled the requirements of ASB Standard 036 in forensic science, which indicated that LPME could be an alternative approach for the investigation of

Table 3
Comparison of reported methods for determination of diphenidol.

Sample	Extraction method	Instrument	Recovery(%)	Linear range	Accuracy(%)	Organic solvent	References
1 mL Plasma or urine	LLE	GC-FID	98	5–200 µg mL ⁻¹	/	4 mL Chloroform	[8]
2 mL Plasma	LLE	HPLC-UV	81	0.04–0.4 µg mL ⁻¹	93–108	4 mL Hexane-ether-isobutanol, (70:25:5 v/v)	[4]
0.1 mL Plasma	LLE	LC-MS-MS	99	0.004–0.4 µg mL ⁻¹	99–103	4 ml Hexane-ether-isobutanol, (70:25:5 v/v)	[10]
0.4 mL Blood	LPME	HPLC-UV	93	1–100 µg mL ⁻¹	90–110	5 µL 2-nonanone	This work
0.4 mL Urine	LPME	HPLC-UV	88	1–100 µg mL ⁻¹	97–109	5 µL 2-nonanone	
40 mg Liver	LPME	HPLC-UV	97	10–1000 µg g ⁻¹	102–108	5 µL NPOE	

biodistribution. Interestingly, the distribution of diphenidol in biofluids was much lower than that in the organs and brain, and the highest concentration of diphenidol was observed in the liver, which is very important for the selection of inspection samples in forensic toxicological analysis. This method shows great potential for the investigation of biodistribution and postmortem redistribution in the fields of forensics.

CRedit authorship contribution statement

Jianhua Xiong: Investigation, Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing – review & editing. **Shuquan Zhao:** Conceptualization, Methodology, Validation, Formal analysis, Writing – review & editing. **Xinwei Zhang:** Methodology, Validation, Formal analysis. **Yu Tian:** Methodology, Validation. **Longda Ma:** Validation, Formal analysis. **Yiwu Zhou:** Conceptualization, Resources, Supervision, Investigation, Methodology. **Chuxiu Huang:** Methodology, Formal analysis, Supervision, Writing – original draft, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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